

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-013 - Nahorkatiya-013

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000146
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2019-02-01 / 2019-04-06
Measured / TVD	2214 m / 2179 m
Trajectory	Directional; inc 35.6 deg; az 171.2 deg
Formation Interval	1371-2170 m
Lithology	Sandstone-siltstone sequence
Pore Pressure Reference	2682 psi
Fracture Gradient	14.63 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (51.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	553	Surface
9 5/8 in casing	1284	Intermediate
7 in liner/casing	1948	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2019-02-14 562 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2019-02-09 395 m	Fishing Operation	High	Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Capture retrieval outcome as institutional knowledge.
2019-02-14 579 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2019-02-19 740 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2019-02-25 961 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2019-03-01 1070 m	NPT	High	Drilling parameters were stable but operations were suspended. Mitigation: Verify equipment health before resuming drilling. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2019-03-12 1420 m	NPT	High	No depth progress occurred during the recorded interval. Mitigation: Verify equipment health before resuming drilling. Lesson: Consistent NPT coding enables meaningful cross-well comparison.

Completion Summary

The synthetic well reached 2214 m MD in the Tipam interval. The NWIS knowledge record links 14 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.